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Predicting rRNA-, RNA-, and DNA-binding proteins from primary structure with support vector machines☆

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Abstract

In the post-genome era, the prediction of protein function is one of the most demanding tasks in the study of bioinformatics. Machine learning methods, such as the support vector machines (SVMs), greatly help to improve the classification of protein function.

In this work, we integrated SVMs, protein sequence amino acid composition, and associated physicochemical properties into the study of nucleic-acid-binding proteins prediction. We developed the binary classifications for rRNA-, RNA-, DNA-binding proteins that play an important role in the control of many cell processes. Each SVM predicts whether a protein belongs to rRNA-, RNA-, or DNA-binding protein class. Self-consistency and jackknife tests were performed on the protein data sets in which the sequences identity was <25%. Test results show that the accuracies of rRNA-, RNA-, DNA-binding SVMs predictions are ~84%, ~78%, ~72%, respectively. The predictions were also performed on the ambiguous and negative data set. The results demonstrate that the predicted scores of proteins in the ambiguous data set by RNA- and DNA-binding SVM models were distributed around zero, while most proteins in the negative data set were predicted as negative scores by all three SVMs. The score distributions agree well with the prior knowledge of those proteins and show the effectiveness of sequence associated physicochemical properties in the protein function prediction. The software is available from the author upon request.

Keywords: rRNA-binding protein; RNA-binding protein; DNA-binding protein; Protein function prediction; Support vector machines (SVMs)

1. Introduction

Large amount of gene sequences have been rapidly produced in genome sequencing projects. The annotation of these enormous gene sequences has been the most important and demanding task within the field of bioinformatics in the post-genome era. By using homology-based protein function annotation methods, only 40–60% of the 35 000–50 000 genes believed to be present in the human genome can be assigned a functional role (Venter et al., 2001; Jensen et al., 2002). So great efforts have been made to develop computational tools for protein function prediction, which include the method based on sequence similarity (Baxevanis, 1998; Bork and Koonin, 1998), evolutionary analysis (Eisen, 1998; Pellegrini et al., 1999), structure (Teichmann et al., 2001), as well as the Rosetta Stone method (Marcotte et al., 1999a), gene fusion (Enright et al., 1999), and gene neighbor approach (Dandekar et al., 1998; Overbeek et al., 1999).

If no clear sequential or structural similarity can be detected between the query protein and one in the annotated database, it will be difficult to identify the function of the query protein by homology based protein function annotation methods. Furthermore, not all

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homologous proteins perform similar functions (Benner et al., 2000). Proteins that contain a shared domain do not necessarily have the same function (Henikoff et al., 1997). For example, many so called "promiscuous" domains such as SH3 are found in various proteins that serve different functions (Marcotte et al., 1999a). To overcome these problems, different approaches were explored, including the improvement in sequence clustering algorithm (Enright et al., 2002) and the combination of existing method (Marcotte et al., 1999b).

A machine learning method, support vector machines (SVMs), is also available for protein function prediction. SVMs are a supervised learning method. They take a set of features, called feature vectors, as inputs to train a model, and output a classification for query by the model. After being trained by a set of feature vectors, whose expected outputs are already known, they are able to classify the later inputting vectors. As a pattern classification engine (Vapnik, 1995), they have been employed in a wide range of studies, including image recognition (Je et al., 2003), text categorization (Kaufman, 1999; Platt, 1999), hand-written digit recognition (Vapnik, 1995), tone recognition (Thubthong and Kijsirikul, 2001), microarray gene expression data analysis (Brown et al., 2000; Furey et al., 2000), and drug design (Burbidge et al., 2000). Recently SVMs have been increasingly used in addressing the problems of protein classification, which include fold recognition (Ding and Dubchak, 2001), subcellular localization prediction (Hua and Sun, 2001; Chou and Cai, 2002), protein structural class prediction (Cai et al., 2001), protein–protein interaction (Bock and Gough, 2001), membrane protein type recognition (Cai et al., 2003c, 2004), G-protein coupled receptors classification (Karchin et al., 2002), and functional classification (Cai et al., 2003a; Cai and Doig, 2004). The prediction results showed that SVMs are comparable or superior to other existing machine learning methods in handling those problems. Thus SVMs may be utilized to solve protein classification problems and to complement the methods based on sequence similarity (Cai et al., 2003a).

Nucleic-acid-binding proteins have a vital role in many aspects of cell function in an organism. For example, DNA-binding proteins play a central role in transcription, packaging, rearrangement, replication and repair (Luscombe and Thornton, 2002). RNA-binding proteins interact with various RNAs at different stages of protein synthesis to control the process of protein synthesis. In this work, we implemented SVMs in the prediction of nucleic-acid-binding proteins. We developed three SVM prototypes to identify whether a query protein belonged to one of three categories, the rRNA-, RNA-, or DNA-binding protein categories. We used protein sequence amino acid composition and a series of associated physicochemical properties to form the feature vectors for SVMs. The properties include hydrophobicity, predicted secondary structure, predicted solvent accessibility, normalized Van Der Waals volume, polarity, and polarizability. Consequently, both distantly and closely related proteins can be predicted (Cai et al., 2003a). We applied the approach on the protein data sets in which the protein identity is less than 25%. The results proved the validity of this method. Greater performance of this method could become possible with further experimentation with various feature vector schemes (Chou, 1999; Chou, 2001).

2. Materials and methods

2.1. Data sets

We used "rRNA-binding", "RNA-binding", and "DNA-binding" as keywords to search the Swiss-Prot database (version 44.0) (Boeckmann et al., 2003; Bairoch et al., 2004), and retrieved 3341, 4819, and 10 476 proteins, respectively. These three collections were designated as "positive" data sets. Searching Swiss-Prot with a list of keywords suspect of implying RNA/DNA-binding functionality, which were proposed by Cai and Lin (2003), a "contrast" data set of 34 938 proteins was retrieved. In order to get the "negative" data set, we removed the proteins in the "contrast" data set from the Swiss-Prot database and obtained 118 933 proteins. Removing the three "positive" data sets from the "contrast" data set, we got an "ambiguous" data set of 18 663 proteins. Proteins in the "ambiguous" data set are probable to have nucleic acid binding functions.

For the calculation of physicochemical properties, protein sequences with length >6000aa or <50aa were moved. Proteins containing irregular amino acid characters such as "x" and "z" were also removed. Then the "ambiguous" data set was reduced to 18 061 proteins. The "negative" data set was reduced to 113 852 proteins. Moreover, redundancy among sequences in "positive" and "negative" data sets was removed by using CD-HIT (Li et al., 2001) and PISCES (Wang and Dunbrack, 2003) program, with a threshold of 25%. It produced 88, 377, 1153, and 17 779 proteins in non-redundant rRNA-, RNA-, DNA-binding and "negative" data sets, respectively. To achieve data balance, we built our data sets in the following manner: first we selected all the proteins in each "positive" subset; and then we randomly selected the same number of proteins from the "negative" data set; then we combined them together. Consequently, three function-specific training data sets were formed, i.e. the rRNA-binding training data set of 176 proteins, the RNA-binding training data set of 754 proteins, and the DNA-binding training data set of 2306 proteins. The non-redundant rRNA-, RNA-, DNA-binding protein data set, the "negative" data set, and the "ambiguous" data set used in this paper are available in Appendix A.

2.2. SVMs

SVMs are a machine learning method based on rigorous statistical learning theories. They were first developed by

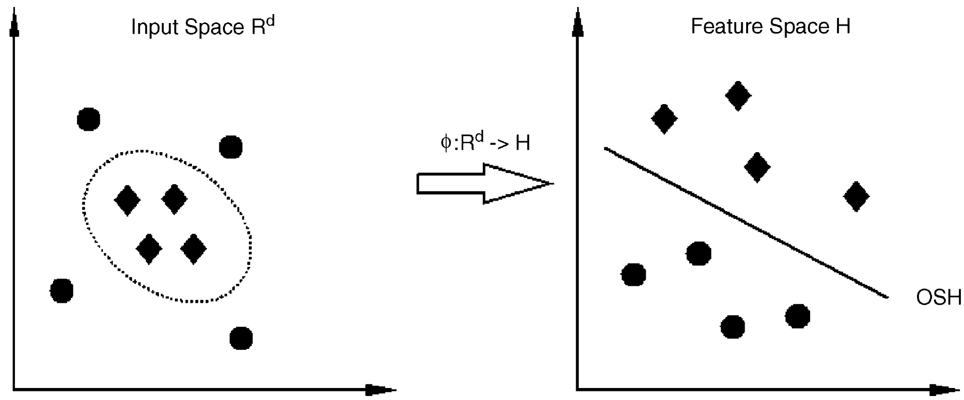

Fig. 1. The basic idea of SVMs: transform the data in the training data set from the input space (R^d) nonlinearly into a higher dimensional feature space (H) via ϕ, where the boundary can be represented by a linear optimal separation hyperplane (OSH). Employing kernel function makes it possible to compute the separating hyperplane without explicitly carrying out the map into the feature space.

Vapnik and colleagues at Bell Laboratories (Vapnik, 1995; Burges, 1998), and then improved by others (Osuna et al., 1997; Joachims, 1998). SVMs can be briefly described as follows. First, SVMs are trained by a group of $\{+1, -1\}$ labeled data formulated as feature vectors. Secondly, these feature vectors ($x \in R^d$) are mapped into a linear or nonlinear feature space ($\phi(x) \in H$) through a kernel function $k(x_i, x_i)$ which defines an dot product in the space H. Thirdly, SVMs try to construct a hyperplane (known as optimal separating hyperplane, OSH) in the feature space, which optimally separates the positive samples and the negative samples (Fig. 1). The separation defines a boundary that maximizes the margin between the hyperplane and the nearest data points of each class in the space.

Suppose we are given a set of samples, i.e. a series of input vectors $x_i \in R^d (i = 1, \ldots, N)$ with corresponding labels $y_i \in \{+1, -1\} (i = 1, \ldots, N)$, where -1 and $+1$ are used to represent the two classes. The objective is to construct a binary classifier or derive a decision function that has small probability of misclassifying a future sample (from the available samples). Both the basic linear separable case and the most commonly used linear non-separable case for real life problems are considered here.

2.2.1. The linear separable case

In this case, there exists a separating hyperplane whose function is $\vec{w} \cdot \vec{x} + b = 0$, which implies: $y_i(\vec{w} \cdot \vec{x}_i + b) \geqslant 1, \quad i = 1, \ldots, N$.

By minimizing $\frac{1}{2}||\vec{w}||^2$ subject to the above constraint, the SVM approach tries to find the OSH, which separates the training data with maximum margin. Here $||\vec{w}||$ is the Euclidean norm of $\vec{w}$ (Cortes and Vapnik, 1995). The classifier is called the largest margin classifier.

By introducing Lagrange multipliers α_i, the Karush–Kuhn–Tucker (KKT) conditions, and the Wolfe dual theorem of optimization theory, the SVM training procedure amounts to solving the following convex quadratic programming (QP) problem:

$$\text{maximum} \quad \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j \cdot y_i y_j \cdot \vec{x}_i \cdot \vec{x}_j$$

subject to the following two conditions:

$$\alpha_i \geqslant 0,$$

$$\sum_{i=1}^{N} \alpha_i y_i = 0, \quad i = 1, \ldots, N.$$

The solution is a unique globally optimized result that has the following expansion:

$$\vec{w} = \sum_{i=1}^{N} y_i \alpha_i \cdot \vec{x}_i.$$

Only if the corresponding $\alpha_i > 0$, these $\vec{x}_i$ are called support vectors.

The decision function that separates the two classes can be written as

$$f(\vec{x}) = \operatorname{sgn}\left(\sum_{i=1}^{N} y_i \alpha_i \cdot \vec{x} \cdot \vec{x}_i + b \right),$$

where sgn( ) is the given sign function.

2.2.2. The linear non-separable case

Two important techniques used in this case are addressed, respectively, as below:

(i) "soft margin" technique

In order to allow for training errors, Cortes and Vapnik introduced slack variables (Cortes and Vapnik, 1995):

$$\xi_i > 0, \quad i = 1, \ldots, N.$$

The relaxed separation constraint is given as

$$y_i(\vec{w}\cdot\vec{x}_i+b)\geqslant 1-\xi_i,\quad (i=1,\ldots,N).$$

Consequently, the OSH can be found by minimizing $\frac{1}{2}||\vec{w}||^2+C\sum_{i=1}^{N}\xi_i$ instead of $\frac{1}{2}||\vec{w}||^2$, under the above two constraints in Section 2.2.1, where C is a regularization parameter to decide the trade-off between the training error and the margin.

(ii) "kernel substitution" technique
SVMs perform a nonlinear mapping of the input vector x from the input space R^d into a higher dimensional Hilbert space, where the mapping is determined by the kernel function. Then like in case (Section 2.2.1), it finds the OSH in the space H corresponding to a nonlinear boundary in the input space.
Two typical kernel functions are listed below:

$$k(\vec{x}_i,\vec{x}_j)=(\vec{x}_i\cdot\vec{x}_j+1)^d,$$

$$k(\vec{x}_i,\vec{x}_j)=\exp(-r||\vec{x}_i-\vec{x}_j||^2),$$

where the first one is called the *polynomial kernel function of degree d* which can revert to the linear function when $d=1$. The second one is called the radial basic function (RBF) kernel.
Finally, for the selected kernel function, the learning task amounts to solving the following QP problem:

$$\text{maximum}\quad \sum_{i=1}^{N}\alpha_i-\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j\cdot y_iy_j\cdot K(\vec{X}_i\cdot\vec{X}_j)$$
$$\text{subject to}\quad 0\leqslant a_i\leqslant C,$$

$$\sum_{i=1}^{N}\alpha_iy_i=0,\quad i=1,\ldots,N.$$

The decision function can be written as

$$f(\vec{x})=\operatorname{sgn}\left(\sum_{i=1}^{N}y_i\alpha_i\cdot K(\vec{x},\vec{x}_i)+b\right).$$

For a given data set, only the kernel function and the regularity parameter C must be selected to specify an SVM.

If a new feature vector is given as an input, its class could be identified according to the side of the hyperplane it is at. Since they always try to seek global optimum and can effectively avoid over-fitting, SVMs can handle a large number of feature vectors (Hua and Sun, 2001; Cai and Lin, 2003). The global optimum is relative to local optimum. It does not necessarily mean automatically seeking the maximum accuracy of training data, because that depends on different kernel functions and their parameters. Based on the principle of *structural risk minimization*, SVMs provide a principled way to estimate generalization performance via an analytic upper bound on the generalization error. This means that a confidence level may be assigned to the prediction. It also means that over-fitting problems can be alleviated (Bock and Gough, 2001). However, the ability of SVMs to avoid over-fitting is also relative. As mentioned above, feature vectors are mapped into a linear or nonlinear feature space through a kernel function. The over-fitting caused by over-tuned kernel function parameters cannot be avoided by SVMs. In summary, the OSH tries to seek global optimum and avoid over-fitting in feature space. However, the distribution of vectors in feature space is predetermined by kernel functions.

The software SVM^{light} was used to implement SVMs into prediction of rRNA-, RNA-, and DNA-binding proteins (Joachims, 1998). SVM^{light} can be freely downloaded from http://svmlight.joachims.org/ for academic use. Three SVM binary classifiers were constructed to identify whether a protein is an rRNA-, RNA-, or DNA-binding protein, respectively. In this work, SVMs parameters were all set as SVM^{light} default. We employed a linear kernel function and obtained the desirable results. It is well known that the linear kernel function does better in avoiding over-fitting than nonlinear function. In order to get a more applicable model for real predictions, we chose linear kernel function to train the models. For more details, readers can consult SVM^{light} specification (Joachims, 1998) and Vapnik's monographs (Cortes and Vapnik, 1995; Vapnik, 1995).

2.3. Feature vector

Constructing an effective feature vectors to represent a protein is the key step for successful SVMs classification. In this research, for every protein sequence, the feature vector was assembled from encoded representations of sequence amino acid composition and tabulated residue properties including hydrophobicity, predicted secondary structure, predicted solvent accessibility, normalized Van Der Waals volume, polarity, and polarizability. The use of the features was motivated by previous studies of proteins (Bock and Gough, 2001; Cai et al., 2003a, b; Ahmad et al., 2004). For hydrophobicity, amino acids were classified into three groups: Arg, Lys, Glu, Asp, Gln, and Asn as polar; Gly, Ala, Ser, Thr, Pro, His, and Tyr as neutral; and Cys, Val, Leu, Ile, Met, Phe, and Trp as hydrophobic (Chothia and Finkelstein, 1990). A three-state model of helix, strand, and coil predicted by Predator was used to represent secondary structure (Frishman and Argos, 1997). The solvent accessibility, predicted by PredAcc, identified each residue as either buried or exposed (Mucchielli-Giorgi et al., 1999). For normalized Van Der Waal volume, polarity, and polarizability, amino acid residues were divided into three groups according to their value range, respectively, (Dubchak et al., 1999).

Three descriptors, composition (C), transition (T), and distribution (D), were used to describe the global composition of each protein properties (Dubchak et al., 1995). C is the percent of amino acids of certain properties. T is the frequency of which amino acids of a certain property is followed by amino acids of a different property. D measures the percent of sequence length within which the

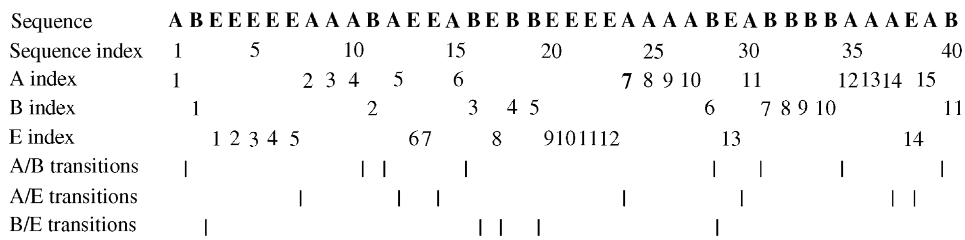

Fig. 2. Hypothetical sequence consisting of three types of residues—A, B, and E—as an example to illustrate derivation of feature vectors (see text for discussion on the sequence descriptors).

first, 25%, 50%, 75% and 100% of the amino acids of a certain property, such as polar, is located, respectively.

A hypothetical protein sequence with 40 amino acids, as shown in Fig. 2, consists of 15 A type residues, 11 B type residues, and 14 E type residues. The first descriptor C is $15/40 = 37.5\%$ for As, $11/40 = 27.5\%$ for Bs, and $14/40 = 35.0\%$ for Es. For the second descriptor T, there are eight transitions between A and B, seven between A and E, and five between B and E. Then the frequencies of these transitions are $8/39 = 20.5\%$, $7/39 = 17.9\%$, and $5/39 = 12.8\%$, respectively. The first, 25%, 50%, 75% and 100% of As are located within 1, 10, 25, 30, and 39 residues, respectively. Thus, the third descriptor D for As is $1/40 = 2.5\%$, $10/40 = 25.0\%$, $25/40 = 62.5\%$, $30/40 = 75.0\%$ and $39/40 = 97.5\%$. Likewise, the D descriptor for Bs is 5.0%, 37.5%, 72.5%, 82.5%, 100.0%, and for Es is 7.5%, 10.0%, 35.0%, 60.0%, 95.0%. Overall, the sequence descriptors for this protein sequence property are C = (37.5%, 27.5%, 35.0%), T = (20.5%, 17.9%, 12.8%), and D = (2.5%, 25.0%, 62.5%, 75.0%, 97.5%, 5.0%, 37.5%, 72.5%, 82.5%, 100.0%, 7.5%, 10.0%, 35.0%, 60.0%, 95.0%). Results show that the solvent accessibility consists of seven-dimensional vectors and other properties each consists of 21-dimensional vectors, respectively. The amino acid composition consists of 20-dimensional vectors (Table 1). Thus, a protein sequence is represented by a 132-dimensional feature vector. This kind of descriptors were first proposed by Dubchak et al. Readers can find more detailed description of this approach in their previous paper (Dubchak et al., 1995).

After the feature vectors were constructed, normalizations were performed among each dimension of vectors in the training data set to adjust the values of all the feature vectors to a standard level. The normalization function is

$$x_{ni} = \frac{x_i - \bar{x}}{s_x},$$

where $s_x = \sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2/n - 1}$, which is the sample standard deviation of x, and $\bar{x} = \sum_{i=1}^{n} x_i/n$.

2.4. Accuracy measure

The quantities of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are often used in the assessment of discriminative methods (Baldi et al., 2000). In this paper, we also used sensitivity, specificity, and overall accuracy to measure prediction (Cai et al., 2003b).

Table 1
Physicochemical feature vectors and their dimensionality

Property name	Dimensions of vector			Total
	C	T	D	
Hydrophobicity	3	3	15	21
Secondary structure	3	3	15	21
Solvent accessibility	1	1	5	7
Normalized Van Der Waals volume	3	3	15	21
Polarity	3	3	15	21
Polarizability	3	3	15	21
Amino acids composition		20		20

Sensitivity, specificity and accuracy of the performance are defined as follows:

Sensitivity = TP/(TP + FN),

Specificity = TN/(TN + FP),

Accuracy = (TP + TN)/(TP + TN + FP + FN).

All these quantities were used in the evaluation of SVM classification performance in this study.

3. Results

SVMs were trained with rRNA-, RNA-, and DNA-binding protein data sets, respectively, to build three corresponding models. As mentioned above, each data set consisted of a positive subset and a negative subset. Proteins in the positive subset were known to have the function that the SVMs were trained to recognize. Proteins in the negative subset were known not to have that function. All the protein functions were determined by their keyword annotations in the Swiss-Prot database, regardless of whether or not the function had been determined experimentally. In order to avoid the bias in selecting negative subsets, we picked five negative subsets randomly from the whole non-redundant negative data set for each of rRNA-, RNA-, and DNA-binding protein training data set. The performance of the SVMs was evaluated by the average accuracy, specificity, and sensitivity from these five data sets. We conducted two tests: self-consistency test and jackknife test, to assess the

Table 2
Self-consistency test results

SVMs	Prediction rate								
	Accuracy (%)			Specificity (%)			Sensitivity (%)		
	Low	High	Average	Low	High	Average	Low	High	Average
rRNA-binding	89.77	94.32	92.84	92.05	95.45	94.09	84.09	94.32	91.59
RNA-binding	81.96	83.55	83.21	76.66	81.70	80.21	85.15	87.27	86.21
DNA-binding	73.46	75.28	74.37	65.39	69.21	66.78	78.06	83.78	81.96

Table 3
Jackknife test results

SVMs	Prediction rate								
	Accuracy (%)			Specificity (%)			Sensitivity (%)		
	Low	High	Average	Low	High	Average	Low	High	Average
rRNA-binding	80.11	87.50	83.98	82.95	88.64	86.82	77.27	87.50	81.14
RNA-binding	74.80	80.24	77.51	69.23	78.51	74.59	77.98	81.96	80.42
DNA-binding	70.08	72.85	71.64	61.32	65.48	63.90	75.72	80.92	79.38

Table 4
CPU time of SVMs training and testing

SVMs	Training (s)	Testing	
		Self-consistency (s)	Jackknife
rRNA-binding	0.14	0.29	51.04 s
RNA-binding	0.69	1.32	16.6 min
DNA-binding	2.52	4.44	2.84 h

performance of each SVM model. In self-consistency test, an SVM was trained and tested with the same data set. After training with all the protein data in the data set, the model was tested with every protein in the training data set by comparing the prediction with known function. For the rRNA-binding protein data set, the SVM model achieved a very good average prediction rate, 92.84% accuracy, 94.09% specificity, and 91.59% sensitivity. For the RNA-binding data set, the SVM model achieved ~83% accuracy, ~80% specificity and ~86% sensitivity. For the DNA-binding data set, the SVM model achieved ~74% accuracy, ~67% specificity and ~82% sensitivity (Table 2).

We adopted the jackknife test for cross-validation. During the jackknife test, every data in the training data set is taken out one by one for prediction, and the remaining data is used to train the model. In other words, the function of each protein is identified by the rules derived from all the other proteins except the one that is being identified. The jackknife test is considered as the most objective and rigorous way to do cross-validation. The success prediction rate in practical application should be measured by the result of jackknife test, but not by arbitrary sub-sampling test or limited independent data set test (Mardia et al., 1979; Zhou and Assa-Munt, 2001). Therefore, in this work, the results got from the jackknife test were considered as the prediction accuracies of the SVMs. They were 83.98%, 77.51%, and 71.64% for the rRNA-, RNA-, and DNA-binding protein data sets, respectively (Table 3). RNA- and DNA-binding proteins are highly diverse both in sequence and in the mode of nucleic acid binding. Some of them may recognize specific RNA/DNA sites, while others are non-specific genome-coating proteins (Cai and Lin, 2003). As compared with them, rRNA-binding proteins are more specialized. Therefore, SVMs achieved the best performance in the rRNA-binding proteins prediction.

We used Dell OptiPlex GX260 computer with an Intel Pentium4 2.40 GHz CPU. The CPU time consumed for training the rRNA-, RNA-, and DNA-binding SVM models were 0.14, 0.69, and 2.52 s, respectively (Table 4). The jackknife test took 51.04 s, 16.6 min, and 2.84 h on the rRNA-, RNA-, and DNA-binding training data sets, respectively. The experiment results indicate that with the increase in the size of training data set, the time cost increased nonlinearly.

4. Discussion

In this study, we employed SVMs to build three binary classifiers to predict rRNA-, RNA-, and DNA-binding proteins. For the different degree of diversity in sequence and functionality, SVMs acquired different prediction rate for rRNA-, RNA-, and DNA-binding proteins. We used self-consistency test and jackknife test to measure the performance of SVM classifiers. In this research, the sequence identity in the protein training data sets was less than 25%. According to the PROSITE database (Falquet et al., 2002), the sequence homology between different rRNA-binding protein families is generally low (Cai and Lin, 2003). The SVMs still achieved a high performance with ~84% accuracy in the jackknife test. The results indicate that the SVMs trained by the amino acid physicochemical properties and sequence amino acid composition have a certain level of capability to classify proteins that are distantly related by sequence. SVMs can find the common factor in a diverse set of positive training data, which the negative data lacks, and use the common factor to find the optimal classification. Thus, this approach may be used as a complementary method to those sequence alignment methods in protein function prediction.

In addition to the self-consistency test and jackknife test performed on the training data sets, we also predicted the proteins in the ambiguous data set and the negative data set. Proteins in the ambiguous data set and the negative data set, which were not used in training the SVM models, were predicted whether they possess one of the three functions with the corresponding SVM model. All the proteins, whose function is known, instead of only those in the non-redundant data sets, were used to train the SVM models. The results demonstrate that when RNA- and DNA-binding SVMs were used, the predicted scores of proteins in the ambiguous data set distributed around zero. The frequency of positive scores and that of negative scores were also balanced (Fig. 3b, c). In addition, most proteins in the negative data set were predicted as negative scores by all three SVMs (Fig. 4a–c). These results agree well with the

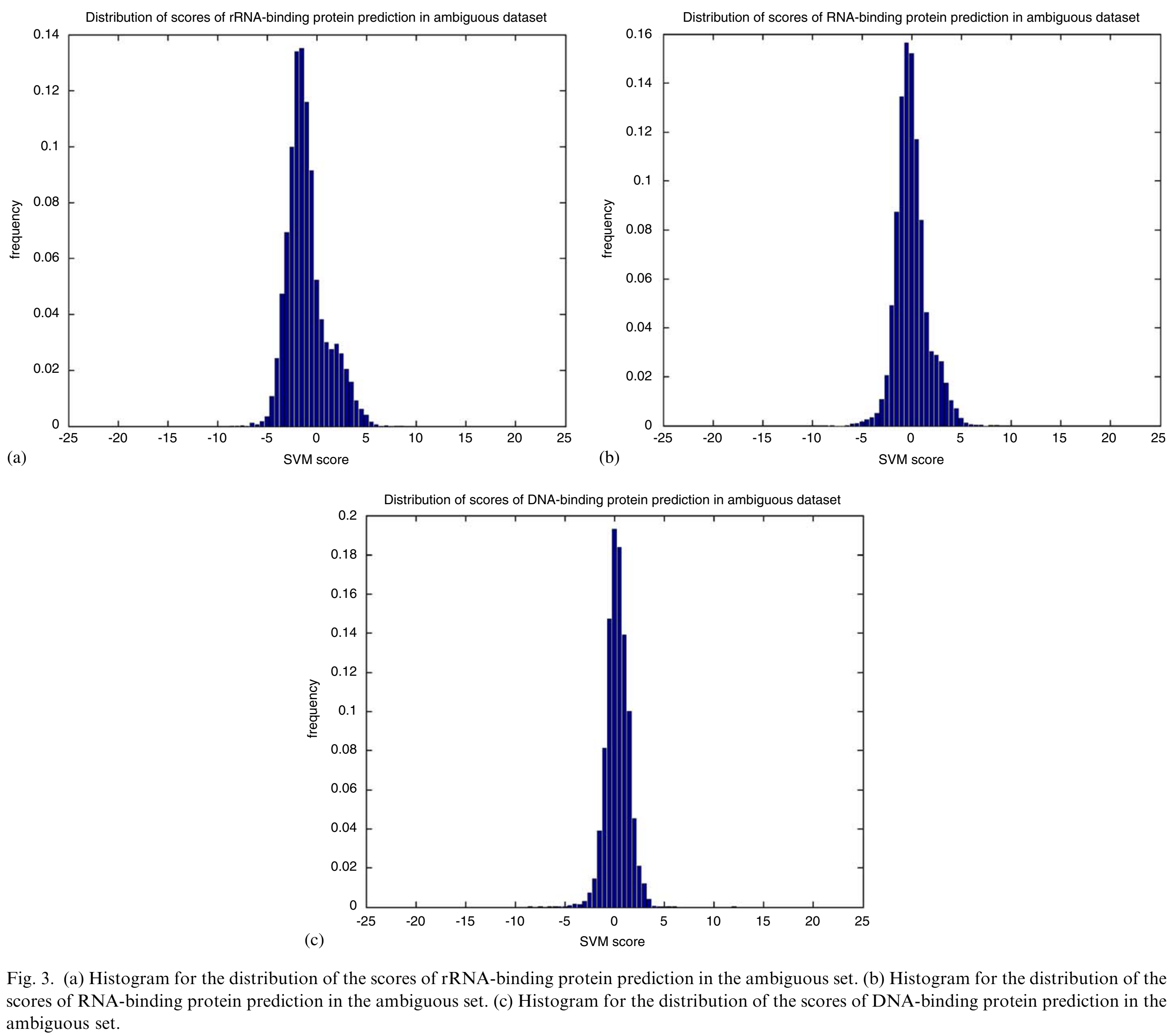

Fig. 3. (a) Histogram for the distribution of the scores of rRNA-binding protein prediction in the ambiguous set. (b) Histogram for the distribution of the scores of RNA-binding protein prediction in the ambiguous set. (c) Histogram for the distribution of the scores of DNA-binding protein prediction in the ambiguous set.

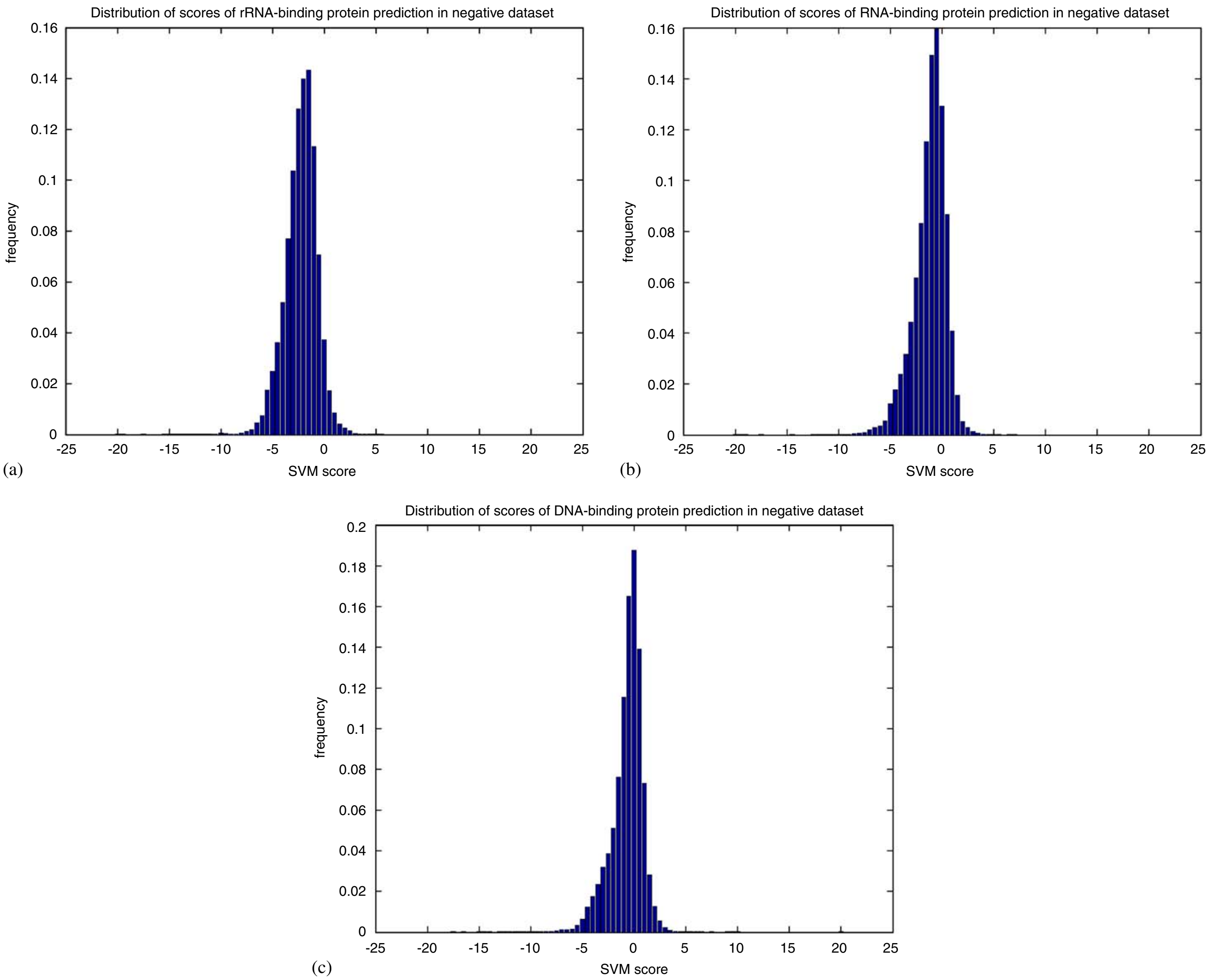

Fig. 4. (a) Histogram for the distribution of the scores of rRNA-binding protein prediction in the negative set. (b) Histogram for the distribution of the scores of RNA-binding protein prediction in the negative set. (c) Histogram for the distribution of the scores of DNA-binding protein prediction in the negative set.

prior knowledge that proteins in the ambiguous data set are suspected to possess nucleic-acids-binding functionality, while proteins in the negative data set are supposed not to have this function. As to the peak deviation of rRNA-binding prediction for the proteins in the ambiguous data set (Fig. 3a), we think the reason may be that the numbers of proteins with rRNA-, RNA-, and DNA-binding function are different in the ambiguous data set. The proteins with rRNA-binding function are much fewer than those with RNA- and DNA-binding functions in Swiss-prot database. There are more DNA- and RNA-binding related keywords in the list used to search for the "contrast" data set (Cai and Lin, 2003). Therefore, we assumed that there were significantly more DNA- and RNA-binding proteins in the ambiguous data set, which might be the reason why most proteins in the ambiguous set were predicted not to possess rRNA-binding function. (Fig. 3a)

When we tried to compare the results with previous studies, we found difficulty in direct comparison, since different data sets, descriptors and classification methods were employed in different work. However, through a tentative comparison, it shows that we achieved similar levels of prediction in terms of accuracy (Cai et al., 2003a, b; Cai and Lin, 2003; Ahmad et al., 2004). Among these studies, Cai and Lin used a similar data set as we did, and got $>95\%$ accuracy in the rRNA-binding protein prediction and various accuracies ranging from $\sim76\%$ to $\sim97\%$ in the RNA- and DNA-binding prediction (Cai and Lin, 2003). However, they did not remove the redundant sequences in their data sets; neither did they perform jackknife test. The accuracies they got were from the

Table 5
Comparison of prediction rate in the DNA-binding protein data sets between vectors calculated from sequence associated physicochemical properties with amino acid composition and vectors calculated only from sequence amino-acid composition

	Prediction rate								
	Accuracy (%)			Specificity (%)			Sensitivity (%)		
	Low	High	Average	Low	High	Average	Low	High	Average
Amino acid composition	68.86	69.99	69.38	63.23	66.96	65.00	71.29	75.46	73.77
Physicochemical properties	70.08	72.85	71.64	61.32	65.48	63.90	75.72	80.92	79.38

ten-time cross-validation test in the redundant data set. Thus, though the accuracies appear to be higher than ours, it does not definitely mean that their method is superior to ours. In order to demonstrate the effectiveness of the feature vectors used in this work, a direct comparison was made. Vectors, generated from sequence associated physicochemical properties and amino acid composition in this work, were compared to the ones calculated only from sequence amino acid composition (Hua and Sun, 2001). As the DNA-binding proteins data set was the biggest data set among rRNA-, RNA-, and DNA-binding proteins data sets, we took the DNA-binding subsets as the representation. The reason we took DNA-binding subsets was also that the proteins in it were the most difficult to predict. The same training and testing processes were preformed with two different feature vectors in the exactly same data sets. The results show that feature vectors calculated from sequence associated physicochemical properties outperformed the vectors calculated only from sequence amino acid composition (Table 5).

In this paper, we did not try other well-known methods for the prediction of secondary structure and solvent accessibility. We realized that the prediction methods we used for secondary structure and solvent accessibility might be trained on proteins homologous to our data set, which may cause some overestimate of our performance. All these elements will be considered in our future research.

In this research, the results show that SVMs, with feature vectors composed of protein sequence amino acid composition and physicochemical properties, can be used to predict nucleic-acid-binding proteins. The success of prediction relies on the low-noise data, effective feature vectors and an appropriate kernel function. Recent work on DNA-binding proteins prediction shows that net charge, electric dipole and quadrupole moment are effective descriptors (Ahmad and Sarai, 2004). This provides us some clues to improve our method in the future study. SVMs attempted to detect the common factor in the nucleic-acid-binding proteins and then employed it to classify proteins. However, presently it is still very difficult to interpret the SVMs internal structure into readable terms. Moreover, researchers may be concerned about its binding sites in addition to a protein's capability of binding nucleic acids. Recently, Ahmad and his colleagues have predicted DNA-binding sites with 40% sensitivity and 79% accuracy by utilizing the artificial neural network (ANN) (Ahmad et al., 2004). Predicting more detailed protein function information by machine-learning methods is also our future direction.

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Appendix A. Supplementary materials

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jtbi.2005.09.018

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